Процедура + Тесты RegisterPlayer

```
CREATE OR ALTER PROCEDURE dbo.RegisterPlayer
    @Nickname    NVARCHAR(50),
    @RealName    NVARCHAR(100),
    @Country     NVARCHAR(50),
    @BirthDate   DATE,
    @TeamId      INT = NULL,
    @NewPlayerId INT OUTPUT
AS
BEGIN
    SET NOCOUNT ON;

    DECLARE @Age INT =
        DATEDIFF(YEAR, @BirthDate, GETDATE())
        - CASE
            WHEN (MONTH(@BirthDate) > MONTH(GETDATE()))
              OR (MONTH(@BirthDate) = MONTH(GETDATE()) AND DAY(@BirthDate) >
DAY(GETDATE()))
            THEN 1 ELSE 0
          END;

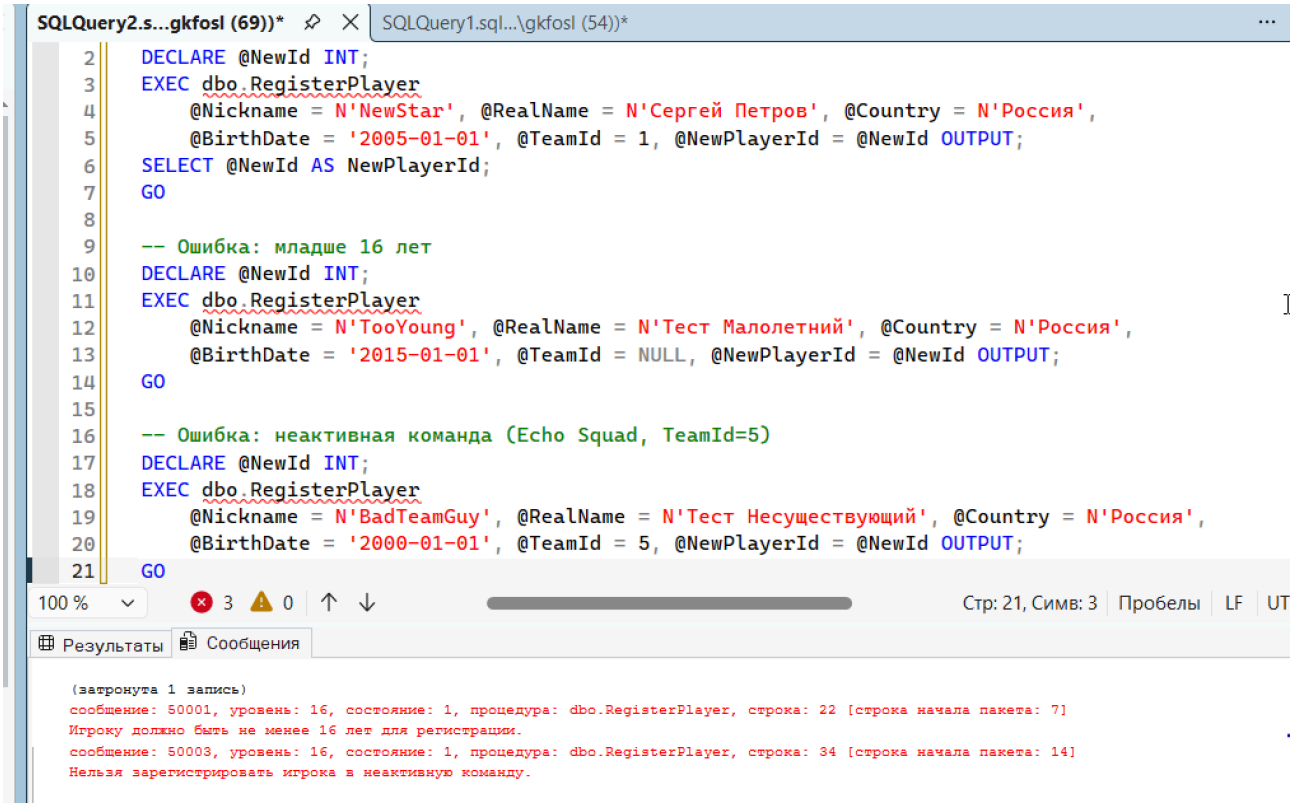
    IF @Age < 16
    BEGIN
        THROW 50001, N'Игроку должно быть не менее 16 лет для регистрации.', 1;
    END

    IF @TeamId IS NOT NULL
    BEGIN
        IF NOT EXISTS (SELECT 1 FROM dbo.Teams WHERE TeamId = @TeamId)
        BEGIN
            THROW 50002, N'Указанная команда не существует.', 1;
        END

        IF EXISTS (SELECT 1 FROM dbo.Teams WHERE TeamId = @TeamId AND IsActive = 0)
        BEGIN
            THROW 50003, N'Нельзя зарегистрировать игрока в неактивную команду.', 1;
        END
    END

    INSERT INTO dbo.Players (Nickname, RealName, Country, BirthDate, TeamId)
    VALUES (@Nickname, @RealName, @Country, @BirthDate, @TeamId);

    SET @NewPlayerId = SCOPE_IDENTITY();
END
GO
```



Процедура + тесты CompleteMatch

```
CREATE OR ALTER PROCEDURE dbo.CompleteMatch
    @MatchId     INT,
    @Team1Score  INT,
    @Team2Score  INT
AS
BEGIN
    SET NOCOUNT ON;
    SET XACT_ABORT ON;

    BEGIN TRY
        BEGIN TRANSACTION;

        IF NOT EXISTS (SELECT 1 FROM dbo.Matches WHERE MatchId = @MatchId)
        BEGIN
            THROW 50010, N'Матч не найден.', 1;
        END

        IF EXISTS (SELECT 1 FROM dbo.Matches WHERE MatchId = @MatchId AND IsCompleted =
1)
        BEGIN
            THROW 50011, N'Матч уже завершён.', 1;
        END

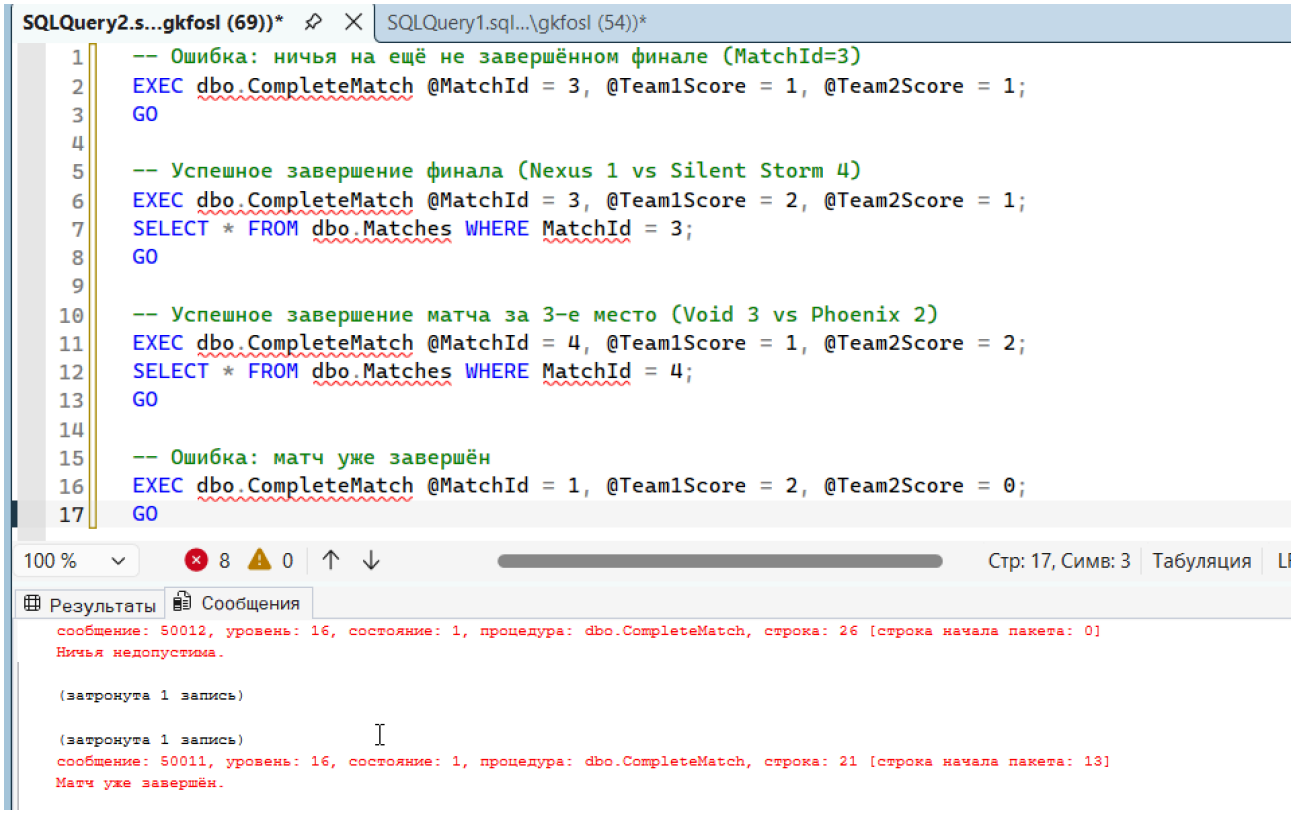
        IF @Team1Score = @Team2Score
        BEGIN
            THROW 50012, N'Ничья недопустима.', 1;
        END

        DECLARE @WinnerId INT;
```

```
        SELECT @WinnerId = CASE WHEN @Team1Score > @Team2Score THEN Team1Id ELSE
Team2Id END
        FROM dbo.Matches
        WHERE MatchId = @MatchId;

        UPDATE dbo.Matches
        SET Team1Score  = @Team1Score,
            Team2Score  = @Team2Score,
            WinnerId    = @WinnerId,
            IsCompleted = 1
        WHERE MatchId = @MatchId;

        COMMIT TRANSACTION;
    END TRY
    BEGIN CATCH
        IF XACT_STATE() <> 0
            ROLLBACK TRANSACTION;
        THROW;
    END CATCH
END
GO
```



Процедура + тесты DistributePrizes

```
CREATE OR ALTER PROCEDURE dbo.DistributePrizes
    @TournamentId INT
AS
BEGIN
    SET NOCOUNT ON;
    SET XACT_ABORT ON;

    BEGIN TRY
        BEGIN TRANSACTION;

        IF NOT EXISTS (
            SELECT 1 FROM dbo.Tournaments
            WHERE TournamentId = @TournamentId AND Status = 'Completed'
        )
        BEGIN
            THROW 50020, N'Турнир должен иметь статус Completed для распределения
призов.', 1;
        END

        IF EXISTS (SELECT 1 FROM dbo.PrizeDistributions WHERE TournamentId = @TournamentId)
        BEGIN
            THROW 50021, N'Призы уже распределены для этого турнира.', 1;
        END

        DECLARE @PrizePool DECIMAL(12,2);
        SELECT @PrizePool = PrizePool FROM dbo.Tournaments WHERE TournamentId =
@TournamentId;

        DECLARE @MaxRound INT;
        SELECT @MaxRound = MAX(Round)
        FROM dbo.Matches
        WHERE TournamentId = @TournamentId AND IsCompleted = 1;

        IF @MaxRound IS NULL
        BEGIN
            THROW 50022, N'Нет завершённых матчей для распределения призов.', 1;
        END

        SELECT MatchId, Round, Team1Id, Team2Id, WinnerId
        INTO #FinalRoundMatches
        FROM dbo.Matches
        WHERE TournamentId = @TournamentId AND IsCompleted = 1 AND Round = @MaxRound;

        SELECT CASE WHEN Team1Id = WinnerId THEN Team2Id ELSE Team1Id END AS LoserId
        INTO #PriorLosers
        FROM dbo.Matches
        WHERE TournamentId = @TournamentId AND IsCompleted = 1 AND Round < @MaxRound;

        DECLARE @FinalWinner INT, @FinalLoser INT, @ThirdWinner INT;

        SELECT TOP 1
            @FinalWinner = WinnerId,
            @FinalLoser  = CASE WHEN Team1Id = WinnerId THEN Team2Id ELSE Team1Id END
        FROM #FinalRoundMatches
        WHERE Team1Id NOT IN (SELECT LoserId FROM #PriorLosers)
          AND Team2Id NOT IN (SELECT LoserId FROM #PriorLosers);
```

```
    IF @FinalWinner IS NULL
    BEGIN
        THROW 50023, N'Не удалось определить финальный матч турнира.', 1;
    END

    SELECT TOP 1 @ThirdWinner = WinnerId
    FROM #FinalRoundMatches
    WHERE Team1Id IN (SELECT LoserId FROM #PriorLosers)
       OR Team2Id IN (SELECT LoserId FROM #PriorLosers);

    DECLARE @Amount1 DECIMAL(12,2) = @PrizePool * 0.5;
    DECLARE @Amount2 DECIMAL(12,2) = @PrizePool * 0.3;
    DECLARE @Amount3 DECIMAL(12,2) = @PrizePool * 0.2;

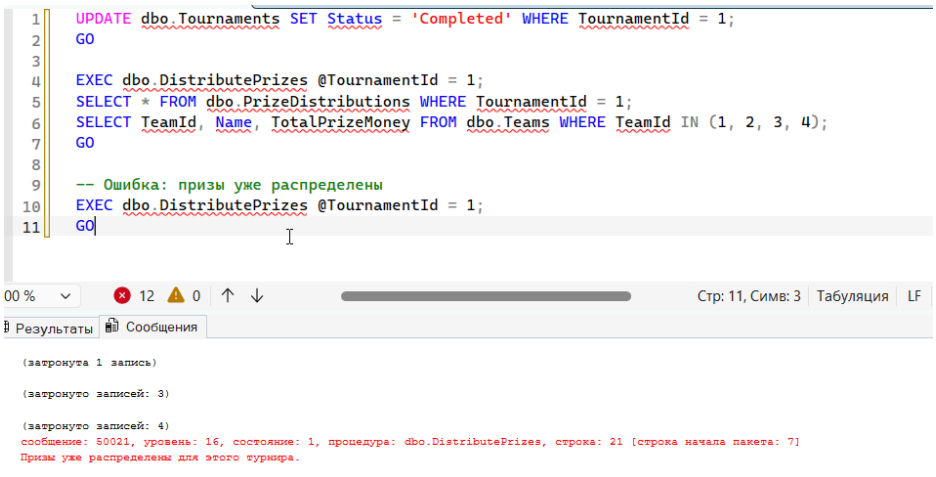
    INSERT INTO dbo.PrizeDistributions (TournamentId, TeamId, Placement, Amount)
    VALUES (@TournamentId, @FinalWinner, 1, @Amount1);
    UPDATE dbo.Teams SET TotalPrizeMoney = TotalPrizeMoney + @Amount1 WHERE TeamId =
@FinalWinner;

    INSERT INTO dbo.PrizeDistributions (TournamentId, TeamId, Placement, Amount)
    VALUES (@TournamentId, @FinalLoser, 2, @Amount2);
    UPDATE dbo.Teams SET TotalPrizeMoney = TotalPrizeMoney + @Amount2 WHERE TeamId =
@FinalLoser;

    IF @ThirdWinner IS NOT NULL
    BEGIN
        INSERT INTO dbo.PrizeDistributions (TournamentId, TeamId, Placement, Amount)
        VALUES (@TournamentId, @ThirdWinner, 3, @Amount3);
        UPDATE dbo.Teams SET TotalPrizeMoney = TotalPrizeMoney + @Amount3 WHERE
TeamId = @ThirdWinner;
    END

    DROP TABLE #FinalRoundMatches;
    DROP TABLE #PriorLosers;

    COMMIT TRANSACTION;
  END TRY
  BEGIN CATCH
    IF XACT_STATE() <> 0
        ROLLBACK TRANSACTION;
    IF OBJECT_ID('tempdb..#FinalRoundMatches') IS NOT NULL DROP TABLE
#FinalRoundMatches;
    IF OBJECT_ID('tempdb..#PriorLosers') IS NOT NULL DROP TABLE #PriorLosers;
    THROW;
  END CATCH
END
GO
```



Функция + тест GetPlayerKDA

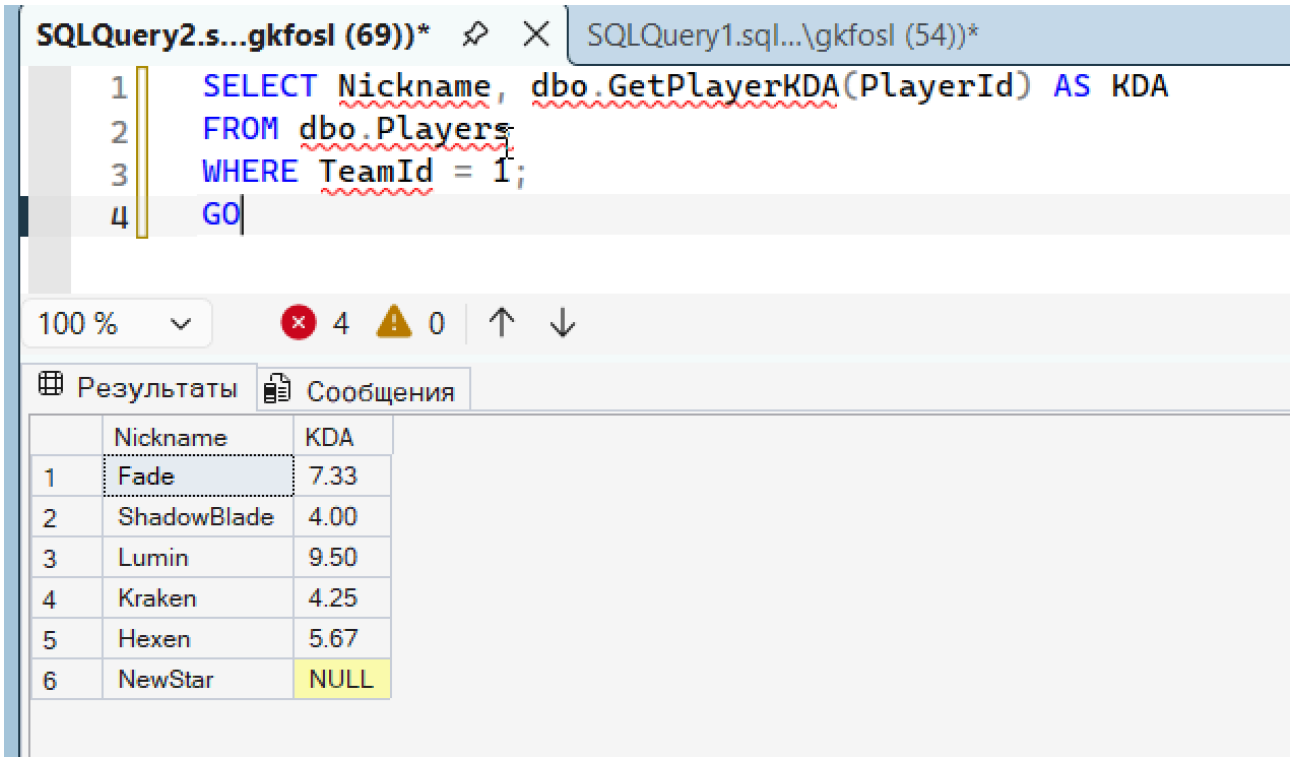
```
CREATE OR ALTER FUNCTION dbo.GetPlayerKDA(@PlayerId INT)
RETURNS DECIMAL(10,2)
AS
BEGIN
    DECLARE @Kills INT, @Deaths INT, @Assists INT, @MatchCount INT;

    SELECT
        @Kills      = SUM(Kills),
        @Deaths     = SUM(Deaths),
        @Assists    = SUM(Assists),
        @MatchCount = COUNT(*)
    FROM dbo.PlayerMatchStats
    WHERE PlayerId = @PlayerId;

    IF @MatchCount IS NULL OR @MatchCount = 0
        RETURN NULL;

    IF @Deaths = 0
        SET @Deaths = 1;

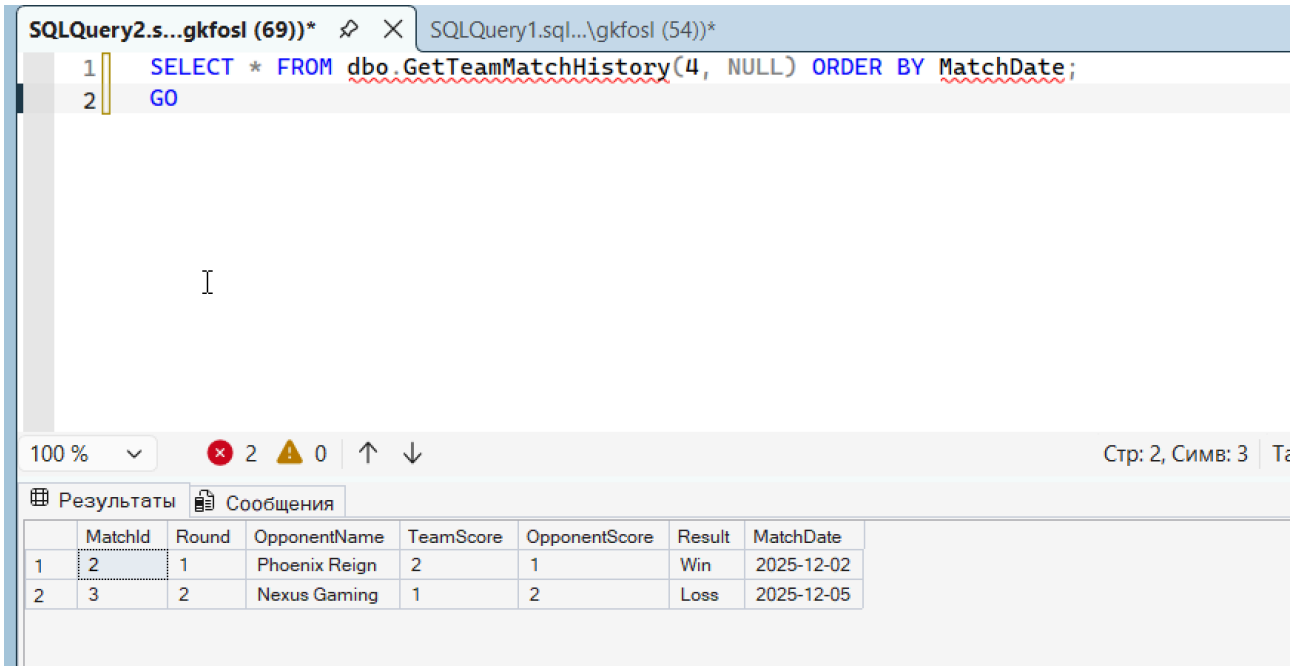
    RETURN CAST((@Kills + @Assists) AS DECIMAL(10,2)) / @Deaths;
END
GO
```



	Nickname	KDA
1	Fade	7.33
2	ShadowBlade	4.00
3	Lumin	9.50
4	Kraken	4.25
5	Hexen	5.67
6	NewStar	NULL

Функция + тест GetTeamMatchHistory (inline TVF)

```
CREATE OR ALTER FUNCTION dbo.GetTeamMatchHistory(@TeamId INT, @GamesBack INT =
NULL)
RETURNS TABLE
AS
RETURN
(
    SELECT TOP (COALESCE(@GamesBack, 2147483647))
        m.MatchId,
        m.Round,
        Opponent.Name AS OpponentName,
        CASE WHEN m.Team1Id = @TeamId THEN m.Team1Score ELSE m.Team2Score END AS
TeamScore,
        CASE WHEN m.Team1Id = @TeamId THEN m.Team2Score ELSE m.Team1Score END AS
OpponentScore,
        CASE
            WHEN m.IsCompleted = 0 THEN 'Pending'
            WHEN m.WinnerId = @TeamId THEN 'Win'
            ELSE 'Loss'
        END AS Result,
        m.MatchDate
    FROM dbo.Matches m
    JOIN dbo.Teams Opponent
        ON Opponent.TeamId = CASE WHEN m.Team1Id = @TeamId THEN m.Team2Id ELSE
m.Team1Id END
    WHERE m.Team1Id = @TeamId OR m.Team2Id = @TeamId
    ORDER BY m.MatchDate DESC
);
GO
```



	MatchId	Round	OpponentName	TeamScore	OpponentScore	Result	MatchDate
1	2	1	Phoenix Reign	2	1	Win	2025-12-02
2	3	2	Nexus Gaming	1	2	Loss	2025-12-05

Функция + тест GetTeamPowerRating (WITH SCHEMABINDING)

```
CREATE OR ALTER FUNCTION dbo.GetTeamPowerRating(@TeamId INT)
RETURNS DECIMAL(18,4)
WITH SCHEMABINDING
AS
BEGIN
    DECLARE @WinRate        DECIMAL(10,4) = 0;
    DECLARE @TotalPrizeMoney DECIMAL(12,2) = 0;
    DECLARE @SponsorBudget   DECIMAL(12,2) = 0;
    DECLARE @Wins INT, @Total INT;

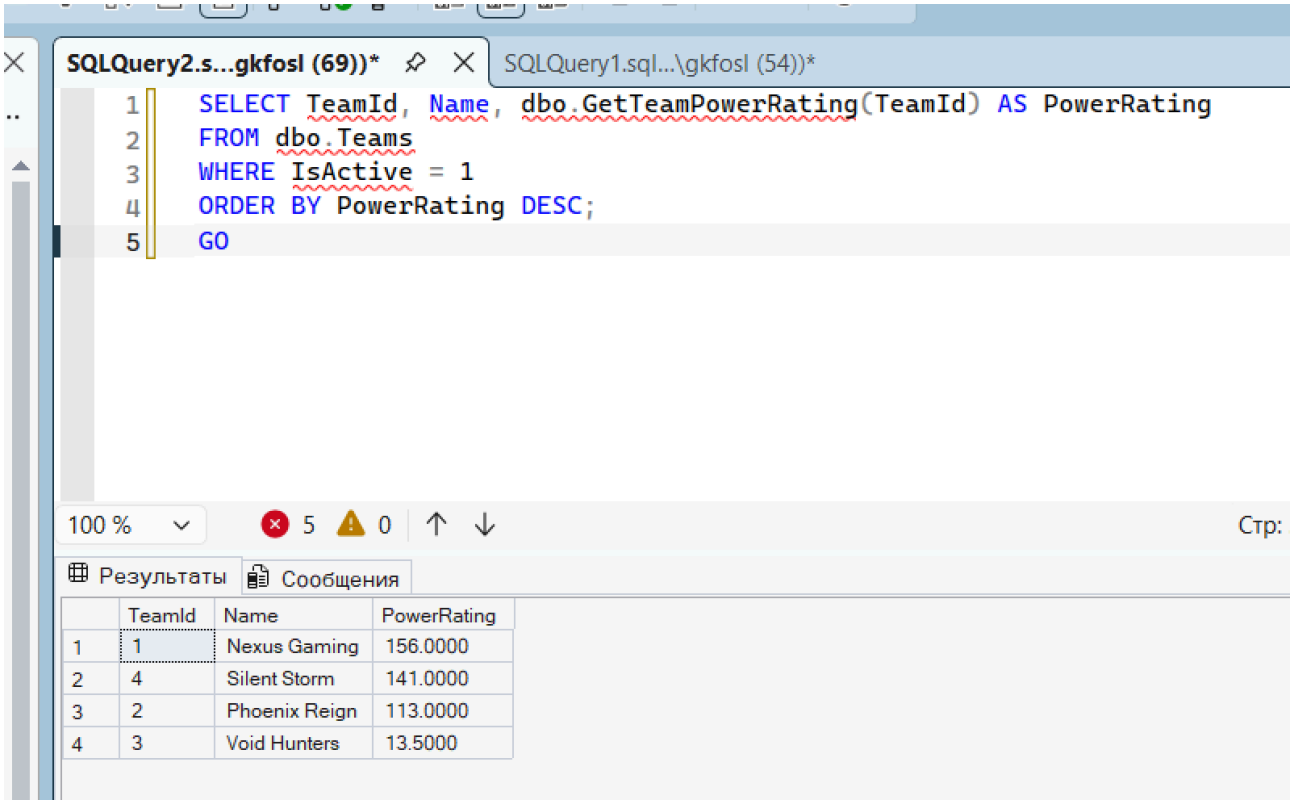
    SELECT
        @Total = COUNT(*),
        @Wins  = SUM(CASE WHEN WinnerId = @TeamId THEN 1 ELSE 0 END)
    FROM dbo.Matches
    WHERE IsCompleted = 1 AND (Team1Id = @TeamId OR Team2Id = @TeamId);

    IF @Total IS NOT NULL AND @Total > 0
        SET @WinRate = CAST(@Wins AS DECIMAL(10,4)) / @Total;

    SELECT @TotalPrizeMoney = TotalPrizeMoney FROM dbo.Teams WHERE TeamId = @TeamId;

    SELECT @SponsorBudget = SUM(ContributionAmount) FROM dbo.TeamSponsors WHERE
TeamId = @TeamId;
    IF @SponsorBudget IS NULL SET @SponsorBudget = 0;

    RETURN (@WinRate * 100) + (@TotalPrizeMoney / 10000) + (@SponsorBudget / 5000);
END
GO
```



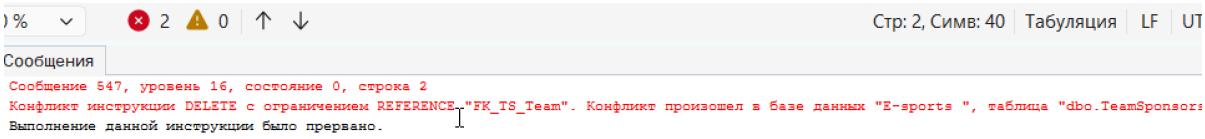
Триггер 1: защита активной команды от удаления

```
CREATE OR ALTER TRIGGER dbo.TR_Teams_PreventDeleteActive
ON dbo.Teams
AFTER DELETE
AS
BEGIN
    SET NOCOUNT ON;

    IF EXISTS (SELECT 1 FROM deleted WHERE IsActive = 1)
    BEGIN
        RAISERROR(N'Нельзя удалить активную команду. Сначала деактивируйте её.', 16, 1);
        ROLLBACK TRANSACTION;
    END
END
GO
```

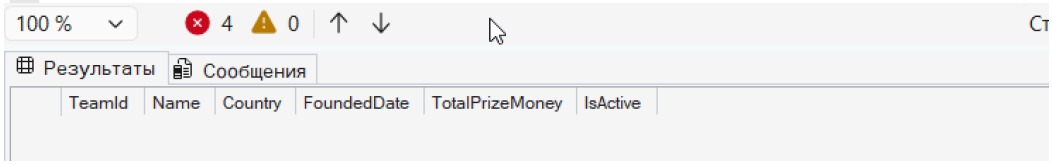
Тест 1

```
1   -- Ошибка: удаление активной команды
2   DELETE FROM dbo.Teams WHERE TeamId = 1;
```



Тест 2

```
1   -- Успешно: удаление неактивной команды (Echo Squad)
2   DELETE FROM dbo.Teams WHERE TeamId = 5;
3   SELECT * FROM dbo.Teams WHERE TeamId = 5; -- должно быть пусто
```



Триггер 2: заглушки статистики при завершении матча

```
CREATE OR ALTER TRIGGER dbo.TR_Matches_AutoCreateStats
ON dbo.Matches
AFTER UPDATE
AS
BEGIN
    SET NOCOUNT ON;

    IF NOT UPDATE(IsCompleted)
        RETURN;

    INSERT INTO dbo.PlayerMatchStats (PlayerId, MatchId, Kills, Deaths, Assists, IsMVP)
    SELECT p.PlayerId, i.MatchId, 0, 0, 0, 0
    FROM inserted i
    JOIN deleted d ON d.MatchId = i.MatchId
    JOIN dbo.Players p ON p.TeamId IN (i.Team1Id, i.Team2Id)
    WHERE d.IsCompleted = 0 AND i.IsCompleted = 1
      AND NOT EXISTS (
          SELECT 1 FROM dbo.PlayerMatchStats pms
          WHERE pms.PlayerId = p.PlayerId AND pms.MatchId = i.MatchId
      );
END
GO
```

View для Триггера 3

```
CREATE OR ALTER VIEW dbo.vw_PlayerTransfers
AS
SELECT
    p.PlayerId,
    p.Nickname,
    p.RealName,
    p.TeamId,
    t.Name AS TeamName
FROM dbo.Players p
LEFT JOIN dbo.Teams t ON t.TeamId = p.TeamId;
GO
```

Триггер 3: INSTEAD OF UPDATE — трансфер игрока

```
CREATE OR ALTER TRIGGER dbo.TR_PlayerTransfers_InsteadOfUpdate
ON dbo.vw_PlayerTransfers
INSTEAD OF UPDATE
AS
BEGIN
    SET NOCOUNT ON;

    IF NOT UPDATE(TeamId)
        RETURN;

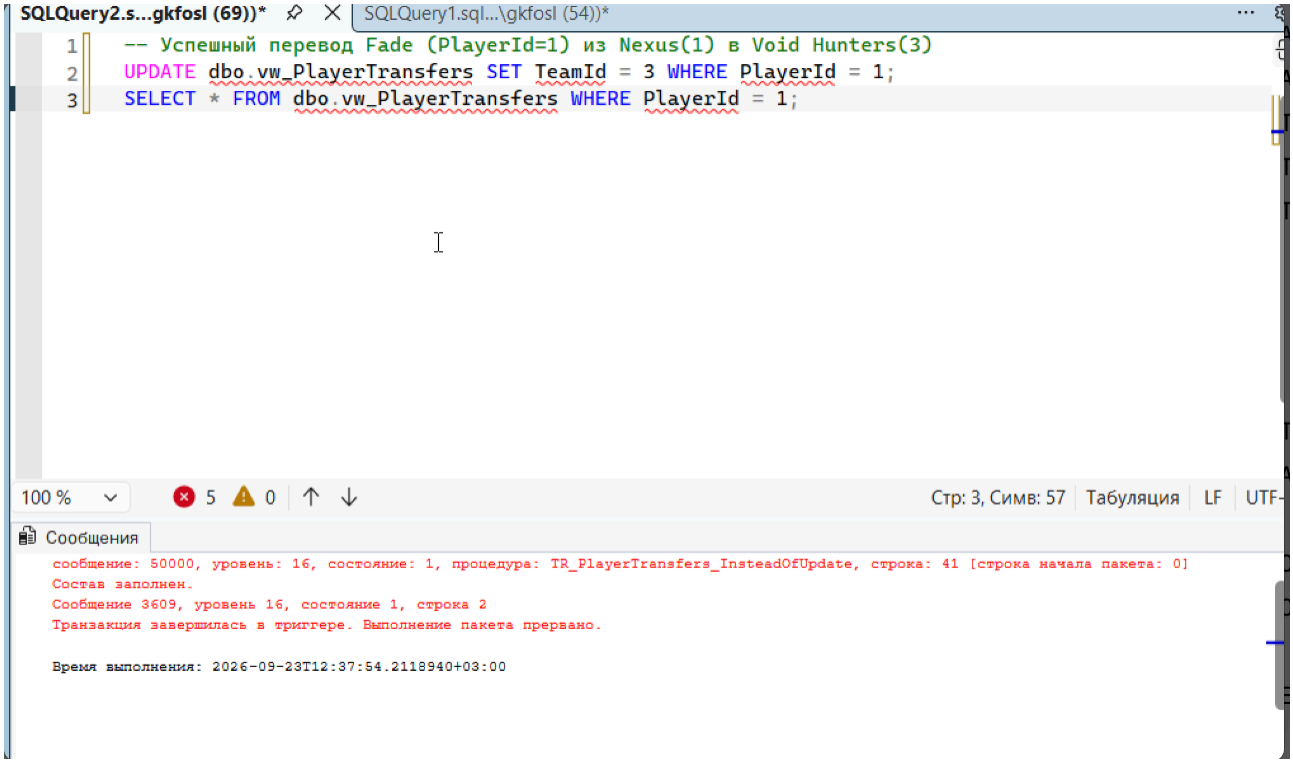
    IF EXISTS (
        SELECT 1 FROM inserted i
        WHERE i.TeamId IS NOT NULL
          AND NOT EXISTS (SELECT 1 FROM dbo.Teams t WHERE t.TeamId = i.TeamId AND
t.IsActive = 1)
    )
    BEGIN
        RAISERROR(N'Целевая команда не существует или неактивна.', 16, 1);
        ROLLBACK TRANSACTION;
        RETURN;
    END

    IF EXISTS (
        SELECT 1 FROM inserted i
        JOIN deleted d ON d.PlayerId = i.PlayerId
        WHERE i.TeamId = d.TeamId
    )
    BEGIN
        RAISERROR(N'Игрок уже состоит в этой команде.', 16, 1);
        ROLLBACK TRANSACTION;
        RETURN;
    END

    IF EXISTS (
        SELECT i.TeamId
        FROM inserted i
        WHERE i.TeamId IS NOT NULL
        GROUP BY i.TeamId
        HAVING (SELECT COUNT(*) FROM dbo.Players p WHERE p.TeamId = i.TeamId) + COUNT(*)
> 5
    )
    BEGIN
        RAISERROR(N'Состав заполнен.', 16, 1);
        ROLLBACK TRANSACTION;
        RETURN;
    END

    UPDATE p
    SET p.TeamId = i.TeamId
    FROM dbo.Players p
    JOIN inserted i ON i.PlayerId = p.PlayerId;
END
GO
```

Тест 1


SQLQuery2.s...gkfosl (69))*
SQLQuery1.sql...\gkfosl (54))*
-- Успешный перевод Fade (PlayerId=1) из Nexus(1) в Void Hunters(3)
UPDATE dbo.vw_PlayerTransfers SET TeamId = 3 WHERE PlayerId = 1;
SELECT * FROM dbo.vw_PlayerTransfers WHERE PlayerId = 1;
100 %
5
0
Стр: 3, Симв: 57
Табуляция
LF
UTF-
Сообщения
сообщение: 50000, уровень: 16, состояние: 1, процедура: TR_PlayerTransfers_InsteadOfUpdate, строка: 41 [строка начала пакета: 0]
Состав заполнен.
Сообщение 3609, уровень 16, состояние 1, строка 2
Транзакция завершилась в триггере. Выполнение пакета прервано.
Время выполнения: 2026-09-23T12:37:54.2118940+03:00

Тест 2


-- Ошибка: перевод в Silent Storm(4), где уже 5 игроков
UPDATE dbo.vw_PlayerTransfers SET TeamId = 4 WHERE PlayerId = 1;
100 %
3
0
Стр: 2, Симв: 65
Табуляция
Сообщения
сообщение: 50000, уровень: 16, состояние: 1, процедура: TR_PlayerTransfers_InsteadOfUpdate, строка: 41 [строка начала пакета: 0]
Состав заполнен.
Сообщение 3609, уровень 16, состояние 1, строка 2
Транзакция завершилась в триггере. Выполнение пакета прервано.